Do Tweets Move Markets?

DS 4002 Case Study by Emily Sun

Every day, financial markets respond to new information: economic reports, global events, and increasingly, direct communication from political leaders. In recent years, public statements from figures like Donald Trump have played a visible role in shaping economic conversations, particularly around trade policy. Tariffs, in particular, have been used as a key policy tool, often announced or discussed through public posts rather than formal channels.

Tariff-related statements can introduce uncertainty into markets by signaling potential changes in international trade relationships, supply chains, and corporate costs. Investors must quickly interpret whether these announcements represent real policy shifts or short-term rhetoric. As a result, even a single post has the potential to influence market expectations and behavior, raising the question of whether these communications have measurable effects on stock performance.

In financial analysis, identifying meaningful signals amid large volumes of data is critical. While most public posts may be unrelated to economic policy, tariff-related messages represent a small but potentially impactful subset. Understanding whether stock returns behave differently on these “event days” compared to normal trading days is essential for evaluating the broader relationship between political communication and market dynamics.

You are a data scientist tasked with determining whether tariff-related posts from Donald Trump are associated with changes in stock market performance. To investigate this, you will analyze a dataset of Trump’s public posts alongside historical S&P 500 data. You will identify tariff-related posts, align them with market trading days, and compare stock behavior on days with and without these announcements. Through statistical analysis and data visualization, you will assess whether these events correspond to meaningful differences in market returns or volatility.